Name: _____

Date: _____

Mr. Rawson • Y9 MYP Physics

Criteria B and C

How every Physics investigation you do is marked

Every lab you write up this year is marked against these two criteria, each **out of 8**. They do not change from experiment to experiment — the springs, the pendulum, and everything after it are all marked on the same two tables. **Keep this sheet.**

Read *down* a column to see how one strand gets better. Read *across* a row to see what a whole level looks like. The green row at the bottom says what each strand means **on a Physics lab**.

You have already done Criterion C once. The spring lab was exactly this: readings in a table, a graph, a line of best fit, and a statement of what the line meant. **Criterion B is the newer one** — it is the planning, and it is marked on work you do *before* you touch the apparatus.

Criterion B — Inquiring and designing *the planning, out of 8*

Level	i · the question	ii · the hypothesis	iii · variables & data	iv · the method
0	The student does not reach a standard identified by any of the descriptors below.			
1–2	states a problem or question to be tested, **with limited success**	states a testable hypothesis	states the variables	designs a method, **with limited success**
3–4	**states** a problem or question to be tested	**outlines** a testable hypothesis using **scientific reasoning**	**outlines** how to manipulate the variables, and states how **relevant** data will be collected	designs a **safe** method, selecting materials and equipment
5–6	**outlines** a problem or question to be tested	outlines **and explains** a testable hypothesis using scientific reasoning	outlines how to manipulate the variables, and **outlines** how **sufficient,** relevant data will be collected	designs a **complete** and safe method, selecting **appropriate** materials and equipment
7–8	**describes** a problem or question to be tested	outlines and explains a testable hypothesis using **correct** scientific reasoning	**describes** how to manipulate the variables, and **describes** how sufficient, relevant data will be collected	designs a **logical**, complete and safe method, selecting appropriate materials and equipment
On a Physics lab	What are you actually trying to find out, and why is it worth knowing?	What do you think will happen — *and why*? Physics, not a guess.	What you change, what you measure, how many readings, and what stays the same.	Numbered steps a classmate could follow without asking you anything. Equipment, and one safety point.

Criterion C — Processing and evaluating *the data, out of 8*

Level	i · presenting data	ii · interpreting	iii · was the hypothesis right?	iv · was the method sound?	v · improvements
0	The student does not reach a standard identified by any of the descriptors below.				
1–2	collects and presents data in numerical and/or visual forms	interprets data	**states** the validity, with **limited reference** to the investigation	**states** the validity, with **limited reference** to the investigation	states **limited** improvements or extensions
3–4	**correctly** collects and presents data	**accurately** interprets data and **describes results**	states the validity, **based on the outcome** of the investigation	states the validity, **based on the outcome** of the investigation	states improvements or extensions **that would benefit** the investigation
5–6	correctly collects, **organizes** and presents data	accurately interprets data and describes results **using scientific reasoning**	**outlines** the validity, based on the outcome	**outlines** the validity, based on the outcome	**outlines** improvements or extensions that would benefit it
7–8	correctly collects, organizes, **transforms** and presents data	accurately interprets data and describes results using **correct** scientific reasoning	**discusses** the validity, based on the outcome	**discusses** the validity, based on the outcome	**describes** improvements or extensions that would benefit it
On a Physics lab	Your results table and your graph, drawn properly, with units.	What the graph shows, said in words, using physics.	Did your own data back the prediction you wrote before you started?	Could your measurements be trusted? Where did the error creep in?	What you would change if you ran it again — and why it would help.

The words that move you up — the strand barely changes between levels, the *verb* does:

states	you wrote it down
outlines	you gave the short version
describes	you said what it is really like
explains	you said *why*
discusses	you weighed it up, both sides

Where the marks actually go. Almost nobody moves up a band by doing *more* experiment. They move up by **writing more carefully about the one they did**.

B is written *before* you touch the apparatus. **C** is written *after*, from your own numbers. Neither is written the night before.